DATA CLEANING TECHNIQUES (EDA & PROBLEM SOLVING)

Used: Microsoft Power BI, Power Query Editor (M language)

Line downtime table

The table was in a wide matrix format, which made analyzing downtime by batch and downtime factor more difficult.

1. Promoted the first row to headers

The actual column names were stored in the first row of the dataset, so we used **Use First Row as Headers** to make them the table's column headers.

2. Unpivoted the downtime factor columns

We kept Batch as the identifier and selected **Unpivot Other Columns**.
This transformed the table from a wide format, where each downtime factor had its own column, into a long format, where each row represents a batch-factor combination.

3. Renamed the columns

- Attribute → FactorID
- Value → Downtime Duration

This made the column names meaningful and easier to understand during analysis.

4. Changed the data type of FactorID

We converted FactorID from **Text** to **Whole Number** because its values represent downtime factor IDs, such as 1, 2, and 7, rather than descriptive text.

Downtime factors table & Products table

Applied the **Trim** transformation to the text columns in both tables to remove leading and trailing whitespace and ensure consistent text values.

Line Productivity table

1. Correcting the data types of time columns

The Start Time and End Time columns initially had the Date/Time data type, which included an incorrect default date (12/31/1899). Both columns were changed to the **Time** data type to retain only the actual time values and remove the incorrect date component.

2. Combining dates and times

Using Power Query's M language, the Date column was combined with Start Time and End Time to create the Start DateTime and End DateTime columns.

3. Handling overnight batches

For batches that continued past midnight, one day was added to the production date used for End DateTime whenever End Time was earlier than Start Time.
This ensured that the end timestamp was chronologically correct and that overnight production durations were calculated accurately.

4. Calculating production duration

A Production Duration column was created by subtracting Start DateTime from End DateTime and converting the resulting duration into minutes.
The column will be used in the analysis phase to evaluate batch production performance.